

... equations in both static and dynamic cases? ✓

What does the phase between the electric and magnetic fields depend on
 $E(r,t) = \vec{E}_0 e^{i(kr - \omega t)}$ / $H(r,t) = \vec{H}_0 e^{i(kr - \omega t - \phi)}$
Write an equation for ϕ

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→ the phase difference depends on the time difference.

the phase difference between the electric and magnetic fields is 180° .

so, it not depends on time
and frequency.

so, it depends on medium.

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According to Faraday's law, the condition of the induced voltage across a terminal of a conducting loop is

.A dynamic magnetic flux linkages the loop .a ☐

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Question

.....Phenomena of creation dipoles in a dielectric material under effect of an applied electric field is called

الاجابة

Power density of an EM radiation is proportional with the relative permittivity of the surrounded medium

اختر واحداً:

☐ صحيح

☐ خطأ

According to Faraday's law, the condition of the induced voltage across a terminal of a conducting loop is

.A dynamic magnetic flux linkages the loop .a ☐

.An electric field is applied on the loop .b ☐

.An electric field surrounds the loop .c ☐

Expert answer

A Anonymous

1) Answer: Polarization. 2) True/Correct

3) Answer is a) Dynamic magnetic flux linkages

What is the uniform plane wave (UPW) meaning?

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The uniform plane wave (UPW) is meaning that all electric and magnetic components are orthogonal on each other.

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What is the polarization of an electromagnetic wave referring to

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* The polarization E.M.W referring to direction of the Electric field of the wave
in E.M.W
E (Electric field) & B (Magnetic field)
are always $\perp$ to each other & $\perp$ to
direction of propagation.

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1.The ac resistance of a slab conductor is depending on ----- and -----.

2.Media are classified as lossless, low-loss, quasi-conducting, or good-conducting on the basis of the ratio---

3.The polarization of an electromagnetic wave referring to direction of the ----- of the -----.

.Answer

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[1]

1) The ac resistance of a slab conductor is depending on 'skin effect' and 'proximity effect'.

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2) Media are classified as lossless, Low loss, quasi-conducting, (or) good conducting on the basis of the ratio Loss Tangent.

$$\checkmark \quad \boxed{\text{Loss Tangent} = \frac{\sigma}{\omega \epsilon}} \left(\frac{J_{\text{conduction}}}{J_{\text{displacement}}} \right)$$

$$\left(\therefore \frac{J_c}{J_D} = \frac{\sigma E}{j\omega \epsilon E} = \frac{\sigma}{j\omega \epsilon} \angle -90^\circ \right)$$

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3) The Polarization of Electromagnetic wave referring to direction of the tip of the Electric field.

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15. How many wavelengths does one complete revolution around a Smith chart represent? $\frac{\lambda}{2}$

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If the physical distance between two points is x on a transmission line, then we need to rotate through an angle of $2\beta x$ degrees on the Smith chart. Because of this, each revolution (one complete revolution) the Smith chart represents a movement of one-half wavelength along the transmission line.

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represent: 

What is the main disadvantage of using coaxial cable at microwave frequencies?

A disadvantage is what kind of filter? High Pass Filter

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The main disadvantage of coaxial cable at microwave region is very high value of attenuation. Hence the main disadvantage of using coaxial cable at microwave frequencies is very high attenuation per foot.

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How many wavelengths does one complete revolution around
What are factors that affect the impedance of a dipole?

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Radiation resistance is a measure of radiated power
for a given i/p current I should be as large as
Possible.

$$= \frac{1}{2} \int_0^\pi \int_0^{2\pi} \left(\frac{60 I_0^2}{r} \cos^2 \left(\frac{\pi}{2} \cos \theta \right) \right) \frac{1}{120\pi} r^2 \sin \theta \, d\phi \, d\theta$$

$$R_r = I_{rms}^2 \cdot 73 \cdot$$

for a half wave dipole = 73 Ω

for a Quarter wave monopole = 36.5 Ω

for a folded dipole  = $2 \cdot 73$
= 146 Ω

In general for n dipole in shunt, $R_r = n^2 \times 73$

- i) What is the effect on transmitted power when load and transmission line are mismatched?
- ii) Under which conditions will all incident power on a line be reflected?

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There are two conditions in which total incident power on a line will be reflected.

- i . When the load end is shorted
- ii. When the load end is opened.

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b. Write the equation

c. The capacitance of a parallel-plate capacitor is depending on
d. What are the factors that affect the capacitance?

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- c. Size of the plates
- d. Separation of the plates
- e. Material filling in the space between the plates

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In a good conductor, the phase of H lead or lag that of E depends on the conductivity.

Select one:

- ☐ True
☒ False

The only effect modeled by an inductor (ideal inductor) in circuit theory is the emf due to self-induction, e_{ind} .

Select one:

- ☐ True
☒ False

According to Faraday's law, the condition of the induced voltage across a terminal of a conducting loop is

- ☐ a. An electric field is applied on the loop.
☐ b. Light linkage the loop.
☒ c. A dynamic magnetic flux linkages the loop.
☐ d. An electric field diverge the loop.

[Clear my choice](#)

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The general shape of the EM wave's polarization is linear.

Select one:

- ☒ True
☐ False

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Power density of an EM radiation is proportional with the relative permittivity of the surrounded medium.

Select one:

- ☒ True
☐ False

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Power density of an EM radiation is proportional with the relative permittivity of the surrounded medium.

Select one:

- ☐ True
☐ False

The direction of the oscillation of an electric field is called

Answer:

Attenuation means that a wave loses energy as it propagates in a lossy medium, where the lost energy is converted to an electromagnetic radiation.

Select one:

- ☐ True
☐ False

Soln

$$① P_{avg} = \frac{P_t h_t}{4\pi d^2} \quad W/m^2$$

Power density (P_{avg})

So P_{avg} doesn't depend on permittivity of medium.

So given statement is false.

② Direction of oscillation of an electric field is called polarization.

$$③ |E| = E_0 e^{-\alpha z}$$

$E \rightarrow$ energy

In lossy medium energy is lost it is not converted in radiation it is lost in medium.
So given statement is false.

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Consider a straight metallic wire of finite length situated in a quasi-static (low-frequency) electromagnetic field. Which of the two quantities, the induced electromotive force (emf), $\mathcal{E}_{ind}$, and current intensity, i , must be zero along the wire?

- ☐ a. $\mathcal{E}_{ind}$ only.
- ☒ b. i only.
- ☐ c. Both quantities.
- ☐ d. Neither of the quantities.

[Clear my choice](#)

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In a good conductor, the phase of H lead or lag that of E depends on the conductivity.

Select one:

☐ True

☒ False

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1. In a good conductor, the phase of H lead or lag that of E depends on the conductivity.

Select one: True False

2. The uniform plane wave (UPW) is meaning that all electric and magnetic components are orthogonal on each other.

Select one: True False

3. Attenuation means that a wave loses energy as it propagates in a lossy medium, where the lost energy is converted to an electromagnetic radiation.

Select one: True False

- ① we know that-
- $$\eta = \frac{E}{H} \text{ and } \theta \eta = \tan^{-1}(\omega \epsilon / \sigma)$$
- For good conductor, $\sigma \gg \omega \epsilon$
 so $\theta \eta = 45^\circ$ $\therefore \tan^{-1}(\infty) = 90^\circ$
- so electric field lead magnetic field by 45° . so H phase will always lag.
 So statement is false.
- ② Uniform plane wave means phase should be constant.
 i.e. $\omega t - \beta z = \text{constant}$
 So statement is false.
- ③ ^{True} False, by attenuation amplitude of electric field decrease with distance travel by wave and it is the loss of power but ~~not~~ an electromagnetic radiation.

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The only effect modeled by an inductor (ideal inductor) in circuit theory is the emf due to self-induction, e_{ind} .

Select one:

- ☐ True
☐ False

True

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- ☐ a. An electric field is applied on the loop.
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- ☒ c. A dynamic magnetic flux linkages the loop.
- ☐ d. An electric field diverge the loop.

Clear my choice

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In a good conductor, the phase of H lead or lag that of E depends on the conductivity.

Select one:

- ☐ True
- ☐ False

The general shape of the EM wave's polarization is linear.

Select one:

- ☐ True
- ☐ False

According to Faraday's law, the condition of the induced voltage across a terminal of a conducting loop is

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Solution:

Q1) for a good conductor Electric field lead magnetic Field always by 45 Degree So H phase will be always Lag

So the statment is False

Q2) the linear polarization or magnetic field vector given plane a long the direction of propogation

So general Shape of the EM Waves polarization is linear

so the Statemnt is True Q3) Faraday Law

$$E = -N \frac{d\Phi}{dt}$$

E is the induced electro magnetic field (EMF) or induced voltage across a conducting loops terminal.

the N is the number of turns of the loop and Φ is the magnetic flux linkages of the loop.

So Option (C) is a Correct answer

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CONCEPTUAL QUESTION 7.48 Shielding effectiveness of an aluminum foil vs. frequency. In order to prevent the electric and magnetic fields from entering or leaving a room, the walls of the room are shielded with a 1-mm-thick aluminum foil. The best protection is achieved at a frequency of

- (A) 1 kHz
- (B) 10 kHz
- (C) 100 kHz
- (D) 1 MHz
- (E) No difference.

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or] as in plane Electromagnetic Waves in lossy Media

$$\alpha = \omega \sqrt{\frac{\epsilon_0 \mu_0}{2}} \left(\sqrt{1 + \left(\frac{\sigma}{\omega \epsilon} \right)^2} - 1 \right)^{1/2}$$

$$\sigma = 1 \quad 1 = \omega \sqrt{\frac{\epsilon_0 \mu_0}{2}} \left(\sqrt{1 + \left(\frac{1}{\omega \epsilon} \right)^2} - 1 \right)^{1/2}$$

$$\left(\frac{1}{\omega \sqrt{\frac{\epsilon_0 \mu_0}{2}}} \right)^2 = \sqrt{1 + \left(\frac{1}{\omega \epsilon} \right)^2} - 1$$

$$\frac{1}{\omega^2} \left(\frac{2}{\epsilon_0 \mu_0} \right) = \sqrt{1 + \left(\frac{1}{\omega \epsilon} \right)^2} - 1$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}, \mu_0 = 4\pi \times 10^{-7} \text{ N/A}^2$$

$$\text{so via} \quad \frac{1}{\omega^2} (1.795 \times 10^{17}) = \left(\sqrt{1 + \frac{10^{12}}{\omega \times 8.854}} - 1 \right)$$

on hit trial we get $\omega = 100 \text{ MHz}$
 1 MHz (approx)

as the Eqⁿ can all be
 solved by hit & trial.

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